

MP2_APIscraping

Celine_Al Miari_and_Yareli_Rosales

```
#| column: page
#| echo: true

#load packages
library(tidyverse)
library(httr)
library(stringr)
library(rvest)
library(rlang)
library(styler)

knitr::opts_chunk$set(
  echo = TRUE,
  message = FALSE,
  warning = FALSE,
  tidy = "styler"
)
```

```
#| column: page
#| echo: true

# Giving to the URL for public MusicBinz API
base_url <- "https://musicbrainz.org/ws/2/artist/"

# The names of artists that we want to study
artists <- c(
  "The Beatles", "Taylor Swift", "Radiohead", "Daft Punk",
  "Miles Davis", "Shawn Mendes", "KAROL G", "Fairuz", "Nancy Ajram"
)
```

Function that will find artists in database

```
artist_finder <- function(artists) {  
  # 1. Set up the Master Tibble  
  music_master <- tibble(  
    Artist = character(),  
    MBID = character(),  
    Type = character(),  
    Gender = character(),  
    Country = character(),  
    Hometown = character(),  
    Start_Date = character(),  
    End_Date = character(),  
    Is_Active = logical(),  
  
    # Quantitative Variables  
    Album_Count = double(),  
    Recording_Count = double(),  
    Work_Count = double(),  
    User_Rating = double(),  
    Rating_Votes = double(),  
  
    # Categorical Variables  
    Top_Genre = character(),  
    Tag_Count = double(),  
    Label_Count = double(),  
    Collab_Count = double()  
  )  
  # 2. The Master For Loop  
  for (i in 1:length(artists)) { # looping around the artists  
  
    # Step A: Find the ID  
    search_url <- str_c(  
      base_url, "?query=",  
      str_replace_all(artists[i], " ", "+"), "&fmt=json"  
    )  
    Sys.sleep(1) # politely scraping  
    # user_agent for Public API  
    search_res <- GET(search_url, user_agent("StOlaf_DataScience_Final"))  
    # get the content of artists parsed  
    search_details <- content(search_res, "parsed")  
    # gets the first artists ID
```

```

artist_id <- search_details$artists[[1]]$id

# Step B: The Lookup
lookup_url <- str_c(
  base_url, artist_id,
  "?inc=release-groups+recordings+works+ratings+tags+artist-rels+label-rels+area-rels+pl
) # looking for the URL for the artists ID
# Sys.sleep(1.1)
details <- content(
  GET(
    lookup_url,
    user_agent("StOlaf_DataScience_Final")
  ),
  "parsed"
) # details artists

# Step C: Filling the Master Row
music_master[i, "Artist"] <- artists[i]
music_master[i, "MBID"] <- artist_id %||% "Unknown"
music_master[i, "Type"] <- details$type %||% "Unknown"
music_master[i, "Gender"] <- details$gender %||% "N/A"
music_master[i, "Country"] <- details$area$name %||% "Unknown"
music_master[i, "Hometown"] <- details$`begin-area`$name %||% "Unknown"
music_master[i, "Start_Date"] <- details$`life-span`$begin %||% "N/A"
music_master[i, "End_Date"] <- details$`life-span`$end %||% "N/A"
music_master[i, "Is_Active"] <- !as.logical(details$`life-span`$ended)

# Quantitative Stats
# API limits to only 25 entities
music_master[i, "Album_Count"] <- length(details$`release-groups`)
music_master[i, "Recording_Count"] <- length(details$recordings)
music_master[i, "Work_Count"] <- length(details$works)
music_master[i, "User_Rating"] <- as.numeric(details$rating$value %||% NA)
music_master[i, "Rating_Votes"] <- as.numeric(details$rating$`votes-count` %||% 0)

# Categorical Counts
music_master[i, "Tag_Count"] <- length(details$tags)
music_master[i, "Top_Genre"] <-
  if (length(details$tags) > 0) details$tags[[1]]$name else NA

# Relationship Counts (Calculating Label and Artist-Collab connections)
if (!is.null(details$relations)) {

```

```

    music_master[i, "Label_Count"] <-
      sum(sapply(details$relations, function(x) x$`target-type` == "label"))
    music_master[i, "Collab_Count"] <-
      sum(sapply(details$relations, function(x) x$`target-type` == "artist"))
  } else {
    music_master[i, "Label_Count"] <- 0
    music_master[i, "Collab_Count"] <- 0
  }
}

# After running the for loop we clean the date column
# to separate between birthdates(person) and start dates(groups)
year <- as.numeric(substr(music_master$Start_Date, 1, 4))
current_year <- 2026

music_master$Age <- ifelse(
  grepl("^\\d{4}-", music_master$Start_Date),
  current_year - year, NA
)

music_master$Start_Date <- ifelse(
  # grepl: a function used to search for pattern matches within a char vector
  grepl("^\\d{4}$", music_master$Start_Date),
  music_master$Start_Date, NA
)

return(music_master)
}

## Call the function on vector given
music_master <- artist_finder(artists)

## Write tibble created into a csv file
write.csv(music_master, "global_music.csv")

```